

3) Estimate J_ε

$$|J_\varepsilon| \leq \int_0^\varepsilon \left(\int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) dy \right) ds$$

$$\sup_{\substack{x \in \mathbb{R}^n \\ t > 0}} \left| \frac{\partial f}{\partial t}(x, t) - \Delta f(x, t) \right|$$

$$= C_\varepsilon \rightarrow 0 \quad (\varepsilon \downarrow 0).$$

4) Compute I_ε :

Note that

$$\frac{\partial f}{\partial t}(x-y, t-s) = \frac{\partial}{\partial t} f(x-y, t-s)$$

$$= -\frac{\partial}{\partial s} f(x-y, t-s)$$

$$\Delta f(x-y, t-s) = \sum_i \frac{\partial}{\partial x_i} \cdot \frac{\partial}{\partial x_i} f(x-y, t-s)$$

$$= \sum_i \frac{\partial}{\partial y_i} \cdot \frac{\partial}{\partial y_i} f(x-y, t-s)$$

$$= \Delta_y (f(x-y, t-s))$$

Then by the integration by parts.

$$\int_\varepsilon^t \int_{\mathbb{R}^n} \frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \left(\frac{\partial f}{\partial t}(x-y, t-s) - \Delta f(x-y, t-s) \right) dy ds$$

$$= \int_\varepsilon^t \int_{\mathbb{R}^n} \left(\frac{\partial}{\partial s} - \Delta_y \right) \left(\frac{1}{(4\pi s)^{n/2}} \exp\left(-\frac{|y|^2}{4s}\right) \right) f(x-y, t-s) dy ds$$

$$- \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y|^2}{4t}\right) f(x-y, 0) dy$$

$= K$

$$+ \int_{\mathbb{R}^n} \frac{1}{(4\pi \varepsilon)^{n/2}} \exp\left(-\frac{|y|^2}{4\varepsilon}\right) f(x-y, t-\varepsilon) dy$$

About Assingment?

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Thus

$$I_\varepsilon = -K + \int_{\mathbb{R}^n} \frac{1}{(4\pi \varepsilon)^{n/2}} \exp\left(-\frac{|y|^2}{4\varepsilon}\right) \cdot f(x-y, t-\varepsilon) dy$$

$$\rightarrow -K + f(x, t) \quad (\varepsilon \downarrow 0) \quad \text{by Thm 12.2}$$

$$\therefore I_\varepsilon + J_\varepsilon + K \rightarrow -K + f(x, t) + K = f(x, t) \quad \square$$

Chapter 14: Maximum principle for heat eq

Thm 14.1

Let $\Omega \subset \mathbb{R}^n$ be a bdd open set.

Let $u: \overline{\Omega} \times [0, T] \rightarrow \mathbb{R}$ be a conti func. up to b'dry
(conti on $\overline{\Omega} \times [0, T]$)

such that u is twice diff'ble w.r.t x and is a diff'ble w.r.t.

If u satisfies

$$\frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, \quad x \in \Omega, t > 0. \quad (14.1)$$

then

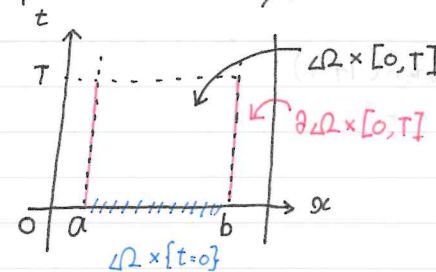
$$\sup_{(x, t) \in \Omega \times [0, T]} u(x, t) = \sup_{(x, t) \in P_B(\Omega \times [0, T])} u(x, t). \quad (14.2)$$

where

$$P_B(\Omega \times [0, T]) := (\partial \Omega \times [0, T]) \cup (\Omega \times \{t=0\})$$

is called "parabolic boundary".

< parabolic boundary >



$$\Omega = (a, b) \in \mathbb{R}$$

pf. 1) Let us first assume

$$(14.3) \quad \frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) < 0, \quad x \in \Omega, t > 0$$

Since u is conti on $\bar{\Omega} \times [0, T]$ u has be assume:
 $\exists (x_0, t_0) \in \bar{\Omega} \times [0, T]$ s.t.

$$u(x_0, t_0) = \max_{(x, t) \in \bar{\Omega} \times [0, T]} u(x, t).$$

Assume contrary $(x_0, t_0) \in \Omega \times [0, T]$.

Then.

$$\frac{\partial u}{\partial x_i}(x_0, t_0) = 0. \quad (i = 1, 2, \dots, n)$$

$$\Delta u(x_0, t_0) \leq 0.$$

$$\frac{\partial u}{\partial t}(x_0, t_0) \geq 0.$$

plugging then into (14.3)

$$\frac{\partial u}{\partial t}(x_0, t_0) - \Delta u(x_0, t_0) < 0$$

and we get contradiction.

Therefore $(x_0, t_0) \in P_B(\Omega \times [0, T])$

$$\Rightarrow \max_{(x, t) \in \bar{\Omega} \times [0, T]} u(x_0, t_0) = \max_{(x, t) \in P_B(\Omega \times [0, T])} u(x, t)$$

hence (14.2) is deduced.

2) We show (14.2) under (14.1)

Let $v(x, t) := u(x, t) - \varepsilon t$

for small $\varepsilon > 0$

Then.

$$\frac{\partial v}{\partial t}(x, t) - \Delta v(x, t)$$

$$= \frac{\partial u}{\partial t}(x, t) - \varepsilon - \Delta u(x, t)$$

$$= -\varepsilon < 0. \quad (\because (14.1))$$

hence by the result of 1).

$$\sup_{(x, t) \in \Omega \times [0, T]} v(x, t) = \sup_{(x, t) \in P_B(\Omega \times [0, T])} v(x, t)$$

For $(x, t) \in \Omega \times [0, T]$

$$u(x, t) = v(x, t) + \varepsilon t$$

$$\leq \sup_{(x, t) \in \Omega \times [0, T]} v(x, t) + \varepsilon T$$

$$= \sup_{(x, t) \in P_B(\Omega \times [0, T])} v(x, t) + \varepsilon T.$$

Note that.

$$u(x, t) = v(x, t) + \varepsilon t \geq v(x, t)$$

we have

$$\sup_{(x, t) \in P_B(\Omega \times [0, T])} u(x, t) \geq \sup_{(x, t) \in P_B(\Omega \times [0, T])} v(x, t)$$

Thus

$$u(x, t) \leq \sup_{(x, t) \in P_B(\Omega \times [0, T])} u(x, t) + \varepsilon T$$

$$\Rightarrow \sup_{(x, t) \in \Omega \times [0, T]} u(x, t) \leq \sup_{(x, t) \in P_B(\Omega \times [0, T])} u(x, t) + \varepsilon T$$

Taking $\varepsilon \downarrow 0$ we have (14.2) $\square$

Next we consider to extend Thm 14.1 to the whole space $\Omega = \mathbb{R}^n$

Thm 14.2

Let $u = u(x, t) : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}$ be a sol of

$$\frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, \quad x \in \mathbb{R}^n, t > 0. \quad (14.10)$$

Assume u is conti on $\mathbb{R}^n \times [0, T]$
twice diff'able w.r.t $x \in \mathbb{R}^n$ a diff'able w.r.t $t > 0$.
Furthermore assume $\exists A, a > 0$ s.t.

$$u(x, t) \leq A e^{a|x|^2}, \quad x \in \mathbb{R}^n, t > 0$$

Then

$$\sup_{(x, t) \in \mathbb{R}^n \times [0, T]} u(x, t) = \sup_{x \in \mathbb{R}^n} u(x, 0)$$

pf. 1) Assume first $4aT < 1$ and take $\varepsilon > 0$ s.t.
 $4a(T + \varepsilon) < 1$.

Fix $y \in \mathbb{R}^n$, $\mu > 0$ and define.

$$v(x, t) := u(x, t) - \frac{\mu}{(T + \varepsilon - t)^{n/2}} \exp\left(\frac{|x - y|^2}{4(T + \varepsilon - t)}\right)$$

Then from (14.10)

$$\frac{\partial v}{\partial t}(x, t) - \Delta v(x, t) = 0, \quad \forall x \in \mathbb{R}^n, 0 < t < T + \varepsilon$$

(Compute Yourself).

For any $r > 0$ from Thm 14.1

we have

$$v(y, t) \leq \sup_{(x, t) \in P_B(B_r(y) \times [0, T - \varepsilon])} v(x, t)$$

2) Let us take

$$(x, t) \in P_B(B_r(y) \times [0, T - \varepsilon])$$

$$\text{If } (x, t) \in B_r(y) \times \{t = 0\}$$

Then

$$v(x, 0) = u(x, 0) - \frac{\mu}{(T + \varepsilon)^{n/2}} \exp\left(\frac{|x - y|^2}{4(T + \varepsilon)}\right) \leq u(x, 0)$$

Else if $(x, t) \in \partial B_r(y) \times [0, T - \varepsilon]$

then $|x - y| = r$, so.

$$v(x, t) = u(x, t) - \frac{\mu}{(T + \varepsilon - t)^{n/2}} \exp\left(\frac{r^2}{4(T + \varepsilon - t)}\right) \leq A \exp(a|x|^2) - \frac{\mu}{(T + \varepsilon - t)^{n/2}} \exp\left(\frac{r^2}{4(T + \varepsilon - t)}\right)$$

Since $4a(T + \varepsilon) < 1$

$$\Leftrightarrow a < \frac{1}{4(T + \varepsilon)}$$

let us put

$$r := \frac{1}{4(T + \varepsilon)} - a$$

Since

$$|x| \leq |x - y| + |y| = r + |y|$$

we have

$$\begin{aligned} v(x, t) &\leq A \exp(a(r + |y|)^2) - \frac{\mu}{(T + \varepsilon - t)^{n/2}} \exp\left(\frac{r^2}{4(T + \varepsilon - t)}\right) \\ &\leq A \exp(a(r + |y|)^2) - \frac{\mu}{(T + \varepsilon)^{n/2}} \exp\left(\frac{r^2}{4(T + \varepsilon)}\right) \\ &\leq A \exp(a(r + |y|)^2) - \mu (4(r + a))^{n/2} \exp(r^2(r + a)) \end{aligned}$$

$\rightarrow$ exercise 14.3.

Since $r > 0$ the RHS of the above goes to $-\infty$ as $r \rightarrow \infty$

Thus for sufficiently large $r > 0$.

$$v(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0) \\ (\alpha \in \partial B_r(y), t > 0)$$

Therefore

$$v(y, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0) \\ (\alpha \in \partial B_r(y), t > 0)$$

Therefore

$$v(y, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0) \\ \therefore v(y, t) \leq \sup_{\alpha \in \mathbb{R}^n} v(\alpha, t) \quad (\alpha, t) \in P(B_r(y) \times [0, T - \varepsilon])$$

Finally taking $\mu \downarrow 0$

$$u(y, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

$\therefore$ If $4aT < 1$ then

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

β) Let us consider general $T > 0$.

First take $T_1 = 1/8a$.

$$\Rightarrow \sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T_1]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Next take $T_2 = 1/4a$

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [T_1, T_2]} u(\alpha, t) = \sup_{\alpha \in \mathbb{R}^n} u(\alpha, T_1) \\ \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

$$\therefore \sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T_2]} u(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$

Repeating the above argument. we have

$$\sup_{(\alpha, t) \in \mathbb{R}^n \times [0, T]} u(\alpha, t) \leq \sup_{\alpha \in \mathbb{R}^n} u(\alpha, 0)$$